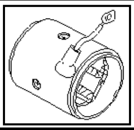
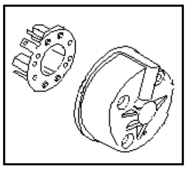
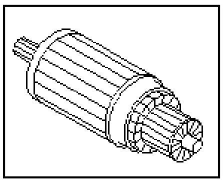
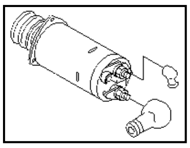
PT SAPTAINDRA SEJATI
NAROGONG REBUILD CENTER

ASSEMBLY CHECK SHEET
STARTING MOTOR

WO NUMBER		PART NAME	STARTING MOTOR	DOCUMENT NO	
ENGINE MODEL		PART NUMBER		PUBLISHED DATE	01/04/2010
ENGINE S/N		START DATE / TIME		REVISION NO	01
MACHINE MODEL		FINISH DATE / TIME		PAGE	1 of 4

ASSEMBLY BY		All dimensions in millimetres All torques in Kilogram metres

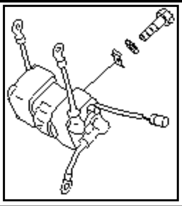
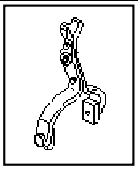
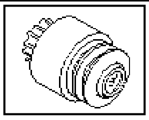
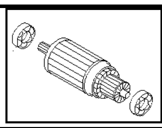
NO.	PARTS	CONDITIONS TO BE CHECKED	SPEC	Sign			REMARKS (SKETCH, PHOTOES, COMMENTS)
				Mec	QA	SPV	
1	Field Coil Ass'y	- Confirm that Yoke Ass'y are free from foreign matter, crack and ovalness - Check o-ring seat. - Check coil by using insulation resistance Tester, confirm that coil doesn't connect with yoke - Check coil from insulation. - Check terminal condition. - Measured field coil ★ Using multi tester. Actual : W ★ Using megger position 500V 1000 M W Actual : W	Min 250 KW Min 250 KW				    
2	Center Bracket Rear Bracket and Front Bracket	- Bracket must free from foreign matters, crack and oval ness - Check contact bearing from ovalness - Check dowel seat and dowel - Check o-ring seat and seal					 
3	Armature	- Armature must free from foreign matter, corrosion, crack and bend. - Check bearing seat measure again / check - Armature by using insulation resistance tester. Actual: W - Confirm that coil does not contact with shaft core, fin /body. - Check and measured diameter contact brushes Actual: mm - Check groove must be free from foreign dust	Min 500 KW std limit 42 mm				 For 170 series, 12V140
4	Magnetic Switch	- Magnetic switch must be free from foreign matter, corrosion, crack and damage - Check all thread hole thread and stud thread - Check magnetic switch function, after repair with condition : Battery 24V - Terminal M – (-) Bat - Terminal C – (+) Bat - Body – (-) Bat - Using multi tester position volt meter measured terminal : B - M	Contact				  

PT SAPTAINDRA SEJATI
NAROGONG REBUILD CENTER

ASSEMBLY CHECK SHEET
STARTING MOTOR

WO NUMBER		PART NAME	STARTING MOTOR	DOCUMENT NO	
ENGINE MODEL		PART NUMBER		PUBLISHED DATE	01/04/2010
ENGINE S/N		START DATE / TIME		REVISION NO	01
MACHINE MODEL		FINISH DATE / TIME		PAGE	1 of 4

ASSEMBLY BY		All dimensions in millimetres All torques in Kilogram metres

NO.	PARTS	CONDITIONS TO BE CHECKED	SPEC	SIGN			REMARKS (SKETCH, PHOTOES, COMMENTS)
				Mec	QA	SPV	
5	Safety Relay Switch	- Safety relay must be free from foreign matter, crack and damage. - Check cables in terminal, confirm that coil. - Check all thread hole thread and stud thread - Check safety relay switch function, by using lamp: - Terminal B_{SR} – (+) Bat - Terminal E_{SR} – (-) Bat - Terminal C_{SR} – (+) lamp - Terminal (-) Bat – (-) lamp - Measured by : - Terminal S_{SR} – (+) Bat - Terminal R_{SR} – (+) Bat	Lamp ON Lamp OFF				 
6	Lever Shift Assy	- Confirm that Lever shift assy is clean, no crack and worn - Check Lever shift assy function. - Install the lever cam - Check position lever code ( ) is true					 
7	Clutch Assy Over Running	- Check it from foreign matter, crack and damage - Check gear from pitting and crack. - Check the rotation (one phase)					 
8	Other Parts	- Check other parts condition, e.g. bearing, seal, shim, o-ring, bolt, washer, bushing, etc.					
9	ASS Armature	- Install bearing in Armature by using adequate tools - Check the bearing rotation.					
10	ASS Motor Assy	- Install Armature in center bracket. - Install o-ring in center bracket that relate with yoke - Install brush in coil with yoke - Install yoke in center bracket and check O-Ring seat - Install Brush in holder brush. - Install holder brush assy on Armature - Install brush on coil yoke to holder brush and check its contact on Armature - Install o-ring on yoke that contact with rear bracket - Install rear bracket - Check mark on yoke and rear bracket - Install o-ring in holder brush bolt and O-Ring in rear bracket bolt - Install bolt in rear bracket than tighten - Test Armature rotation and confirm it. - Test rotation by using 2 x 24V200AH battery - Install cable (-) in body - Install cable (+) in cable terminal (-) - Confirm that motor rotation is smooth.					  

PT SAPTAINDRA SEJATI
NAROGONG REBUILD CENTER

ASSEMBLY CHECK SHEET
STARTING MOTOR

WO NUMBER		PART NAME	STARTING MOTOR	DOCUMENT NO	
ENGINE MODEL		PART NUMBER		PUBLISHED DATE	01/04/2010
ENGINE S/N		START DATE / TIME		REVISION NO	01
MACHINE MODEL		FINISH DATE / TIME		PAGE	1 of 4

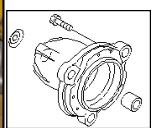
ASSEMBLY BY		All dimensions in millimetres
		All torques in Kilogram metres

PT SAPTAINDRA SEJATI
NAROGONG REBUILD CENTER

ASSEMBLY CHECK SHEET
STARTING MOTOR

WO NUMBER		**PART NAME**	**STARTING MOTOR**	**DOCUMENT NO**	
ENGINE MODEL		**PART NUMBER**		**PUBLISHED DATE**	**01/04/2010**
ENGINE S/N		**START DATE / TIME**		**REVISION NO**	**01**
MACHINE MODEL		**FINISH DATE / TIME**		**PAGE**	**1 of 4**

ASSEMBLY BY		**All dimensions in millimetres** **All torques in Kilogram metres**

NO.	PARTS	CONDITIONS TO BE CHECKED	SPEC	SIGN Mec	SIGN QA	SIGN SPV	REMARKS (SKETCH, PHOTOES, COMMENTS)
11	**Center Bracket Assembly**	- Install oil seal in center bracket A - Install bearing in center shaft - Install shift lever in center bracket A, install seal in shift lever tightening bolt - Install gear clutch over running in bracket center A that linked in lever, and put in shaft center that has been install with bearing (Bearing with LT 2) - Install stopper pinion and put in ring clip.					
12	**Starting Motor Assembling**	- Install shim in center bracket - Install O-Ring in center bracket - Connect center bracket A that has been assembled in motor assy than install bolt and tighten with allen key. **Actual: Kgm** - Check rotation by rotating center shaft. - Install magnetic switch and tighten bolt with L-Shaped **Actual: Kgm** - Confirm that stud adjuster has connected with shift lever. - Install yoke cable (+) in terminal (M) Magnetic switch and tighten nut - Mark cap in magnetic switch - Install safety relay in magnetic switch Note : - Confirm starting motor position (left or right) - Tighten safety relay tightening bolt **Actual: Kgm** - Install cable (B) from Safety Relay to terminal (B) - Magnetic Switch than tighten nut - Install cable (C) from Safety Relay to terminal © - Magnetic Switch and mark cap on connector than thighten nut - Install cable (-) from Safety Relay through mounting bolt magnetic switch and tighten bolt	1.2 ± 0.3 Kgm 0.8 ~ 1.2 Kgm 0.8 ~ 1.2 Kgm				  
13	**Front Bracket**	- Install o-ring in front bracket - Install front bracket assembly in center bracket A - Confirm that seat condition. Note : - Confirm starting motor position (Left or right) - Install bolt and tighten **Actual: Kgm**	0.8 ~ 1.2 Kgm				 

PT SAPTAINDRA SEJATI
NAROGONG REBUILD CENTER

ASSEMBLY CHECK SHEET
STARTING MOTOR

WO NUMBER		PART NAME	STARTING MOTOR	DOCUMENT NO	
ENGINE MODEL		PART NUMBER		PUBLISHED DATE	01/04/2010
ENGINE S/N		START DATE / TIME		REVISION NO	01
MACHINE MODEL		FINISH DATE / TIME		PAGE	1 of 4

ASSEMBLY BY		All dimensions in millimetres
		All torques in Kilogram metres

PT SAPTAINDRA SEJATI
NAROGONG REBUILD CENTER

ASSEMBLY CHECK SHEET
STARTING MOTOR

WO NUMBER		PART NAME	STARTING MOTOR	DOCUMENT NO	
ENGINE MODEL		PART NUMBER		PUBLISHED DATE	01/04/2010
ENGINE S/N		START DATE / TIME		REVISION NO	01
MACHINE MODEL		FINISH DATE / TIME		PAGE	1 of 4

ASSEMBLY BY		All dimensions in millimetres All torques in Kilogram metres

NO.	PARTS	CONDITIONS TO BE CHECKED	SPEC	SIGN			REMARKS (SKETCH, PHOTOES, COMMENTS)
				Mec	QA	SPV	
14	Starting Motor Testing	- Install battery cable (+) on magnetic switch terminal (B). - Install battery cable (-) on starting motor body terminal. - Install safety relay contactor, confirm that the starting motor is good and rotating smoothly					 
15	Final Inspection	- Check all bolts and mark 'X' - Mark JO No. in Starting Motor Body - Replace Starting Motor Name Plate with ***KRI Startng Motor Name Plate.**					

Notes :

ASSEMBLY BY		INSPECTION & APPROVED BY
MECHANIC (MEC)	QUALITY ASSURANCE (QA)	PROD. SUPERVISOR (SPV)